AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM298
Operating Systems

Midterm Examination (MTA)
Spring Semester 2024/2025

Date: 09/Apr./2025

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nber of Exam Pages: 3 uding this cover sheet)	**Time Allowed:** 1.5 Hours

tructions:

- Total Marks: **30**
- This exam consists of **three** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are NOT allowed.

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Part 1:Multiple Choice Questions: [5 marks]

Question 1: Choose the most correct answer.

1. _____ is the time it takes for the dispatcher to stop one process and start another running.
   a. Loader
   b. Dispatch latency
   c. Dispatch loading
   d. Interrupt handling
2. _____ assigns final addresses to program parts and adjusts code and data in program t match those addresses
   a. Relocation.
   b. Loader.
   c. Linker.
   d. Object Oriented.
3. One of the following is not a System Call API :
   a. Win32.
   b. POSIX.
   c. JVM.
   d. Dropbox.
4. Device controller informs CPU that it has finished its operation by causing a/an:
   a. Request
   b. Bus request
   c. Interrupt
   d. Pass
5. Which of the following is an advantage of using threads?
   a. Reduced complexity
   b. Reduced resource utilization
   c. Increased concurrency
   d. Both a and c

Part 2: Short Essay Questions: [15 marks]

Question 2: [2 marks]
What is the main advantage of Multiprogramming? Discuss briefly.

Question 3: [2 marks]
What is a kernel? Define.

Question 4: [3 marks]
We know that a process generally contains multiple parts, three of which are the Text, the Data, and the Stack sections.
What does each of these sections contain?

Question 5: [4 marks]
Explain the difference between data parallelism and task parallelism, providing a figure tha clarifies each type's operation.

Question 6: [4 marks]
What do the following acronyms stand for?
a- DMA
b- CLI
c- PCB
d- GUI

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Part 3: Problem Solving: [10 marks]

Question 7: [4 marks]

a) Write a Cronjob code to run a command called ***Alarm*** every week on Wednesday at 7:30 am. (2 marks)

b) What is the difference between **ps** and **ps -A**? Explain. (2 marks)

Question 8: [6 marks]

Write a Bash script that prompts the user to enter **three** numbers using a loop. Each time, the script should check whether the entered number is **positive, negative, or zero** and display the corresponding message immediately:

- If the number is positive, it displays: "The number is positive."
- If the number is negative, it displays: "The number is negative."
- If the number is zero, it displays: "The number is zero."

NB: Display an empty line after completing each loop for better readability (Check the expected output).

Expected Output:

```
Enter a number: 5
The number is positive.

Enter a number: -15
The number is negative.

Enter a number: 0
The number is zero.
```

End of questions

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